

First Hop Redundancy Protocols

- Topics covered:
 - The Purpose of FHRPs
 - HSRP (Hot Standby Router Protocol)
 - VRRP (Virtual Router Redundancy Protocol)
 - GLBP (Gateway Load Balancing Protocol)
 - Basic HSRP Configuration
-

First Hop Redundancy Protocols (FHRP)

Purpose of First Hop Redundancy Protocols

The purpose of first hop redundancy protocols is to provide redundancy for the default gateway in a network.

The default gateway is the first hop used by a PC when sending traffic to destinations outside its local subnet.

If the default gateway fails, communication to external networks (such as the Internet) will fail unless redundancy is implemented.

Introduction to FHRPs Covered:

Three first hop redundancy protocols are introduced:

- HSRP – Hot Standby Router Protocol
- VRRP – Virtual Router Redundancy Protocol
- GLBP – Gateway Load Balancing Protocol

The goal is basic comparison, not in-depth knowledge.

Later, basic configuration of HSRP will be covered.

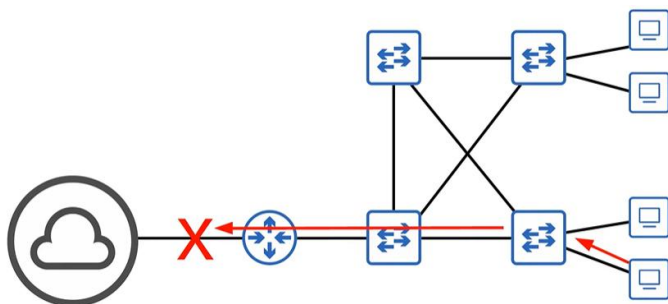
Redundancy Motivation Using a Network Diagram

A network with redundancy is used to demonstrate the importance of having multiple paths.

If there is only one Internet connection and it fails, PCs lose Internet access.

- At home, this is inconvenient.
- For businesses, this is a serious problem.

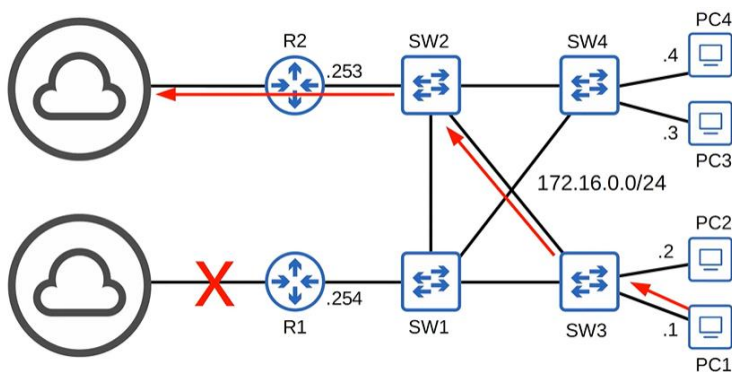
Initial Network Without Redundancy:



- One router connects the network to the Internet.
- Failure of this link stops Internet access completely.

Network With Redundant Internet Connections:

An additional router and Internet connection are added.



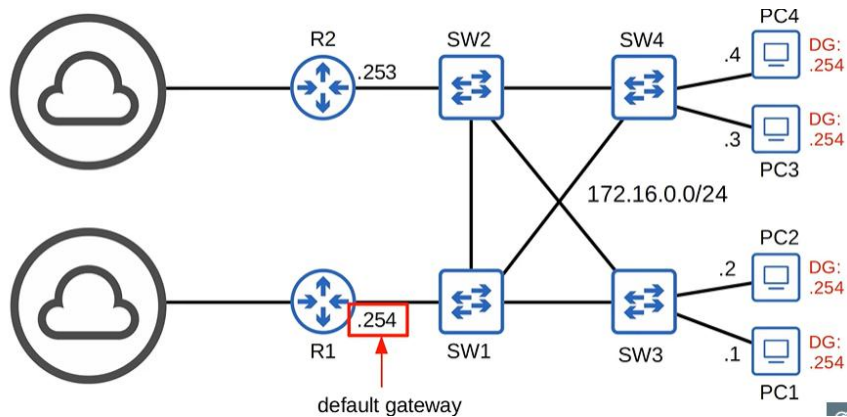
- IP addresses are in the 172.16.0.0/24 network.
- If one connection fails, traffic can use the alternate path.

Default Gateway Problem

R1's IP address 172.16.0.254 is configured as the default gateway on all PCs.

This means:

- Any traffic destined outside the local subnet is sent to R1.



Failure Scenario: R1 Goes Down

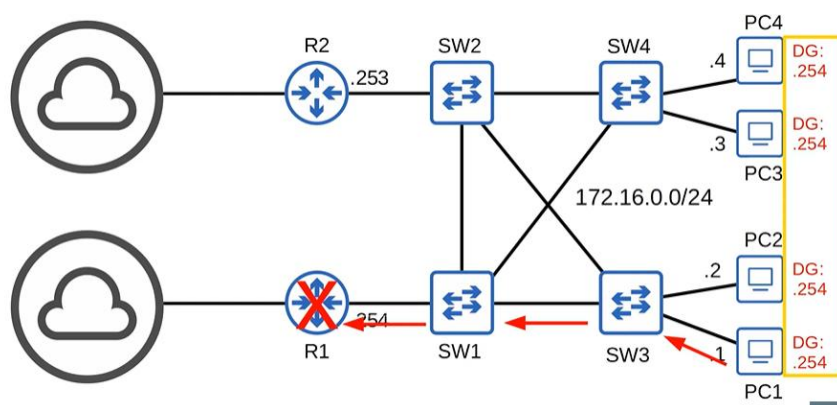
If R1 fails:

- R2 is still connected to the Internet.
- But PCs do not know this.

Problem:

The PCs' default gateway is still set to 172.16.0.254.

They continue sending traffic to R1, even though it is down.



Need for First Hop Redundancy Protocols

The problem: How can R2 automatically become the default gateway when R1 fails?

This is the role of a First Hop Redundancy Protocol (FHRP).

Definition of FHRP:

A first hop redundancy protocol:

- Protects the default gateway
- Allows two or more routers to share a gateway address
- A backup router takes over if the active router fails
- Failover usually occurs within a few seconds

The term first hop refers to the default gateway being the first router in the path.

Virtual IP Address (VIP)

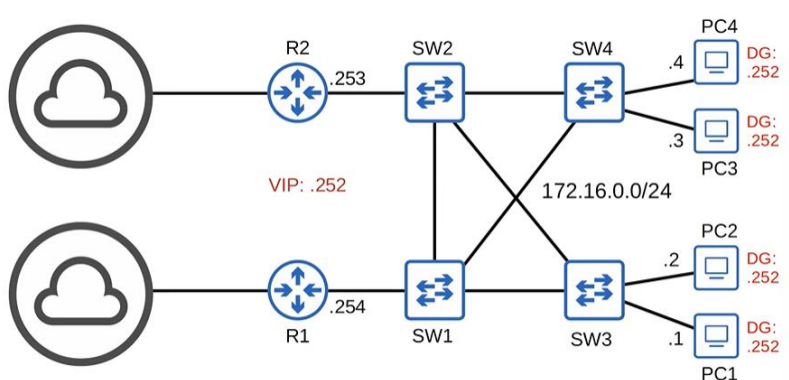
FHRPs use a Virtual IP address (VIP).

Example VIP:

- 172.16.0.252

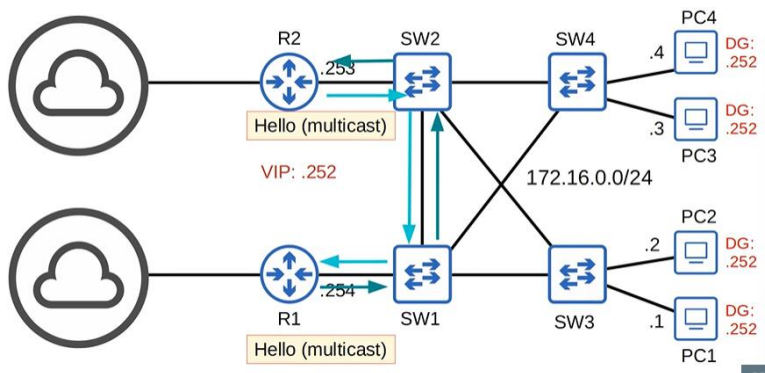
PCs are configured with:

- Default Gateway = 172.16.0.252
- NOT the physical IP of R1 or R2



Router Role Negotiation:

Routers negotiate their roles by sending multicast “Hello” messages.



Active and Standby Roles:

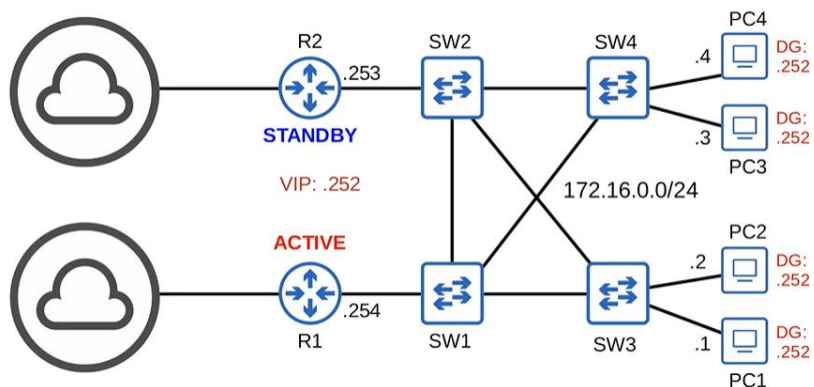
After negotiation:

- R1 becomes the Active router
- R2 becomes the Standby router

Roles:

- Active: Acts as the default gateway
- Standby: Takes over if the active router fails

Role names vary depending on the FHRP used.



Traffic Flow with FHRP Enabled

PC1 needs to send traffic to another network.

PC1's logic:

- Destination is outside the local subnet
- Default gateway = 172.16.0.252
- MAC address is unknown

ARP Request for Virtual IP

PC1 sends an ARP request to learn the MAC address of 172.16.0.252.

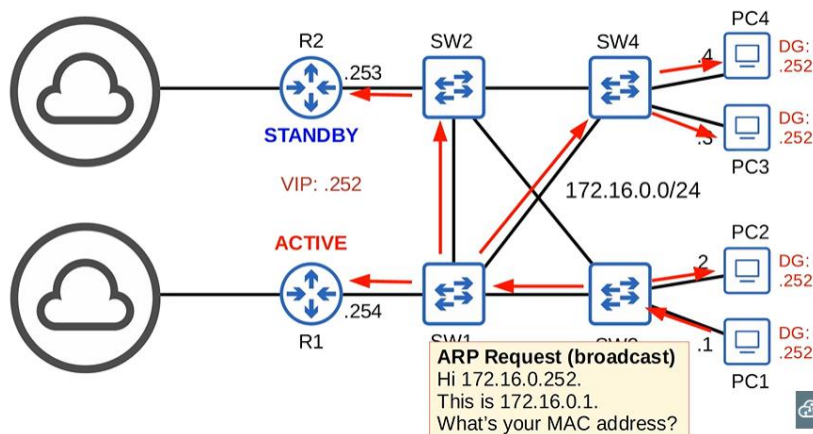
ARP request message:

“Hi 172.16.0.252.

This is 172.16.0.1.

What's your MAC address?”

ARP requests are broadcast.



Spanning Tree prevents broadcast storms by blocking some paths.

ARP Request Reception:

Both R1 and R2 receive the ARP request because:

- Both share the same virtual IP
- The ARP request is broadcast

ARP Reply Decision:

Only the Active router responds.

- R1 = Active
- R2 = Standby

Therefore:

- R1 sends the ARP reply

ARP Reply with Virtual MAC:

R1 sends a unicast ARP reply:

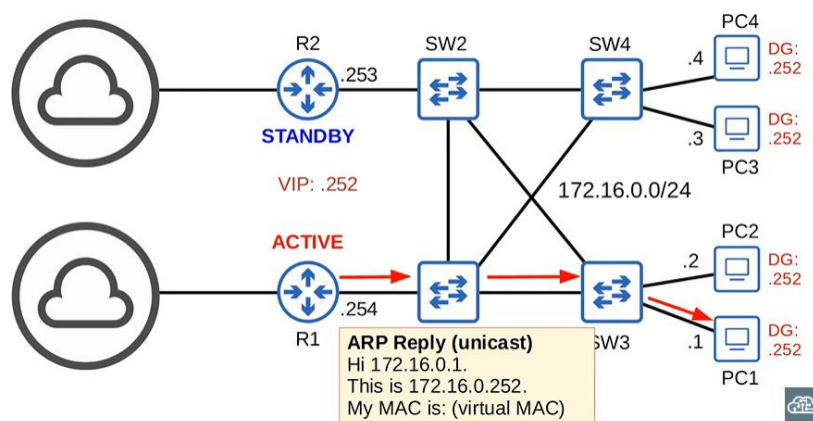
“Hi 172.16.0.1.

This is 172.16.0.252.

My MAC address is (virtual MAC).”

Important:

- FHRPs use a virtual MAC address
- Each FHRP uses a different virtual MAC



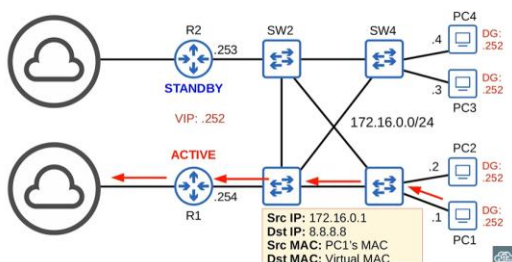
Packet and Frame Fields Example:

PC1 sends traffic to 8.8.8.8.

Fields:

- Source IP: 172.16.0.1
- Destination IP: 8.8.8.8
- Source MAC: PC1's MAC
- Destination MAC: Virtual MAC

The frame is sent to **R1**, and the packet is forwarded to the Internet.



R1 Failure Event

Concept

Now consider the case where R1 goes down.

At this moment:

- There is no active default gateway for the network.
- PCs are still configured with the virtual IP as their default gateway.

Hello Message Timeout and Role Change:

R2 notices that:

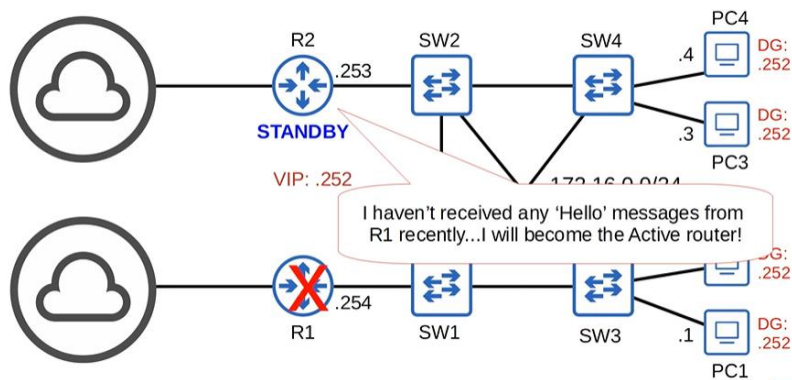
- It has not received Hello messages from R1 for some time.

R2 then decides:

- “I haven’t received any Hello messages from R1 recently.”
- “I will become the Active router.”

So:

- R2 becomes the Active router.



How Traffic Is Redirected to R2:

Question: How does R2 tell other devices to send traffic to it instead of R1?

Important point:

- PCs do NOT need to change their ARP tables.

The PC ARP entry already maps:

- Virtual IP 172.16.0.252
- To the Virtual MAC address

PC ARP Table (Unchanged):

PC1 ARP Table

IP: 172.16.0.252

MAC: Virtual MAC

What Actually Needs to Be Updated:

What must change:

- Switch MAC address tables

Why:

- Switches decide where to forward frames based on their MAC tables.

How Switches Learn MAC Addresses:

Switch behaviour:

- When a switch receives a frame:
 - It checks the source MAC address
 - It updates its MAC address table accordingly

Therefore:

- R2 must send frames using the virtual MAC address as the source MAC.

Gratuitous ARP:

To update switch MAC tables, R2 sends gratuitous ARP messages.

Definition

Gratuitous ARP:

- ARP replies sent without being requested
- No ARP request is received beforehand

Normal ARP:

- ARP reply is sent in response to an ARP request

Gratuitous ARP:

- ARP reply is sent without any request

Broadcast Nature of Gratuitous ARP:

Another difference:

- Gratuitous ARP frames are broadcast
 - Destination MAC: FF:FF:FF:FF:FF:FF
- Normal ARP replies are unicast

Because they are broadcast:

- All switches receive the frames
- All switches update their MAC address tables

Gratuitous ARP Transmission by R2:

R2 sends gratuitous ARP messages stating:

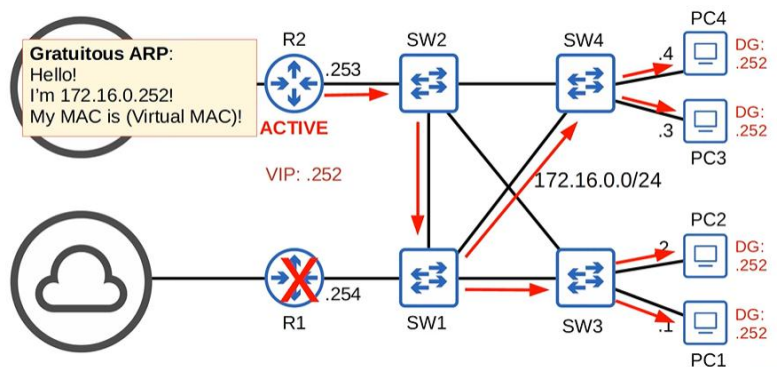
“Hello!

I’m 172.16.0.252!

My MAC is the Virtual MAC!”

These messages are broadcast across the network.

Spanning Tree Protocol prevents a broadcast storm.

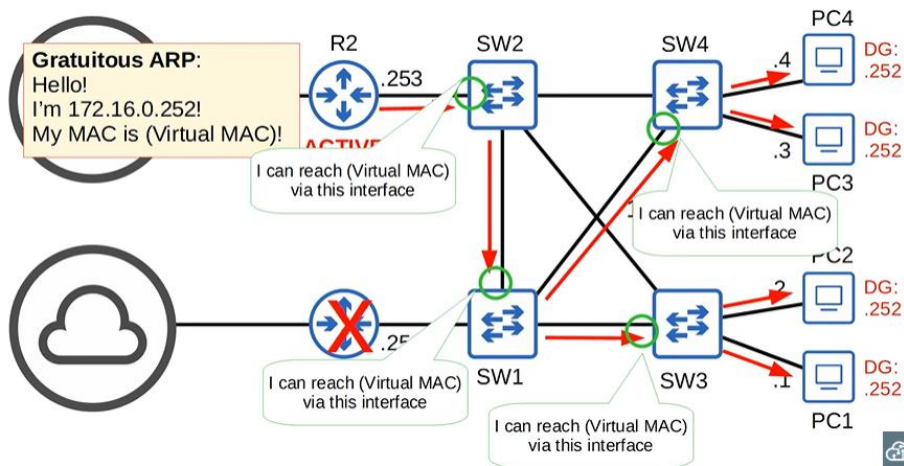


Switch MAC Table Updates:

After receiving the gratuitous ARP frames:

- SW2 learns it can reach the Virtual MAC via R2
- SW1 learns the same
- SW3 learns the same
- SW4 learns the same

All switches now forward frames for the Virtual MAC toward R2.



R2 Fully Active as Default Gateway

At this point:

- R2 is the Active router
- Switch MAC tables are updated
- R2 now functions as the default gateway

Traffic Flow After Failover:

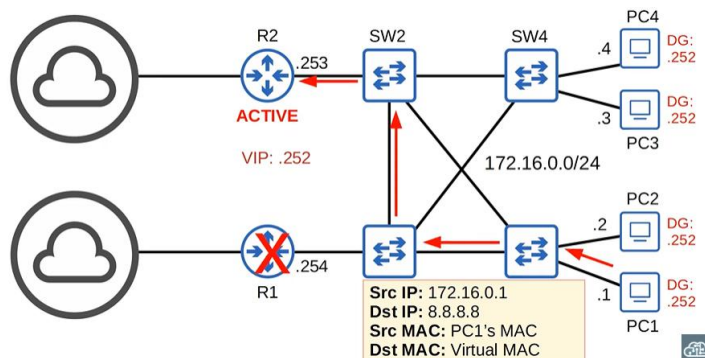
PC1 sends traffic to 8.8.8.8 again.

Important:

- The frame and packet are exactly the same as before

Fields:

- Source IP: 172.16.0.1
- Destination IP: 8.8.8.8
- Source MAC: PC1's MAC
- Destination MAC: Virtual MAC



R1 Comes Back Online

Now consider: R1 comes back online

Question: Will R1 automatically become the Active router again?

Answer: No

Non-Preemptive Behavior of FHRPs:

FHRPs are non-preemptive by default.

Meaning:

- The current active router does not give up its role
- Even if the former active router returns

So:

- R2 remains Active
- R1 becomes Standby

This behaviour is similar to:

- DR and BDR in OSPF

Preemption (Optional Behaviour):

Although non-preemptive is the default:

- You can configure preemption

With preemption enabled:

- R1 will take back the Active role when it comes online
- R2 will become the Standby router again

Summary of FHRP Operation

- A **Virtual IP** is configured on two routers
- A **Virtual MAC** is generated for that Virtual IP
- Each FHRP uses a different Virtual MAC format
- An **Active** and **Standby** router are elected
- End hosts use the **Virtual IP** as their default gateway
- The Active router replies to ARP using the **Virtual MAC**
- Traffic to other networks is sent to the Active router
- If the Active router fails:
 - The Standby becomes Active
 - Gratuitous ARP messages are sent
 - Switch MAC tables are updated
- The new Active router functions as the default gateway
- If the old Active router returns:
 - It becomes Standby by default
- Preemption can be configured to change this behavior

HSRP (Hot Standby Router Protocol)

What HSRP?

HSRP is Cisco proprietary.

- It runs only on Cisco devices
- It cannot run on devices from other vendors (for example, Juniper)

In HSRP:

- One router is elected as Active
- One router is elected as Standby

These are the same terms used earlier when explaining generic FHRP behaviour.

HSRP Versions:

HSRP has two versions

- Version 1
- Version 2

Version 2:

- Adds IPv6 support
- Increases the number of HSRP groups

The concept of an HSRP group will be shown later during configuration.

Multiple Subnets and HSRP Groups:

Earlier examples used:

- One subnet: 172.16.0.0/24
- One VLAN

In networks with:

- Multiple subnets
- Multiple VLANs

Each subnet requires:

- Its own default gateway

- Its own virtual IP address

Each virtual IP address is configured in a separate HSRP group.

HSRP Multicast Addresses:

Routers exchange Hello messages using multicast.

HSRP uses:

- 224.0.0.2 → Version 1
- 224.0.0.102 → Version 2

These addresses are important to remember for the exam.

HSRP Virtual MAC Address Formats:

These virtual MAC formats must be memorized for the exam.

HSRP Version 1 Virtual MAC:

Format:

0000.0c07.acXX

- XX = HSRP group number (hex)

Example:

- Group number = 1
- Virtual MAC = 0000.0c07.ac01

HSRP Version 2 Virtual MAC:

Format:

0000.0c9f.fXXX

- XXX = HSRP group number (hex)
- Three hexadecimal digits allow more groups

Example:

- Group number = 1
- Virtual MAC = 0000.0c9f.f001

Load Balancing Using Multiple HSRP Groups

HSRP cannot load balance within a single subnet.

However:

- Traffic can be load balanced between subnets
- This is done by configuring different active routers per subnet/VLAN

This concept is briefly demonstrated.

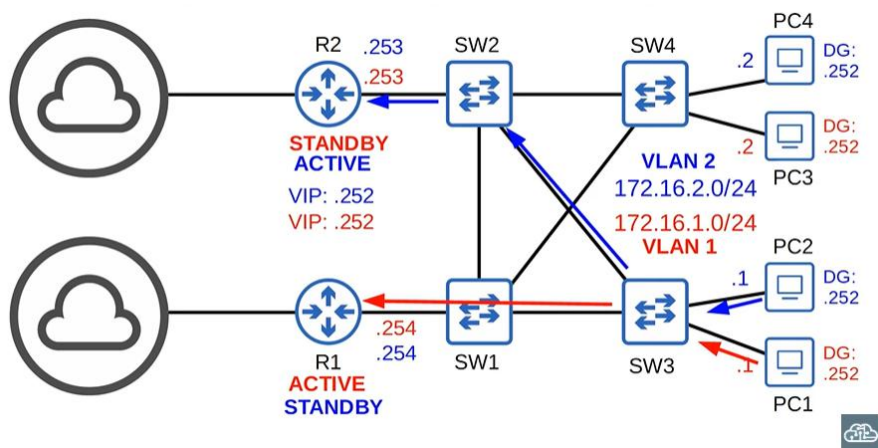
HSRP Load Balancing Example with VLANs:

VLAN and Subnet Assignment

- VLAN 1 → 172.16.1.0/24
 - Hosts: PC1, PC3
- VLAN 2 → 172.16.2.0/24
 - Hosts: PC2, PC4

Each VLAN has:

- Its own virtual IP
- Its own HSRP group



Summary of HSRP Capabilities:

- One Active router per subnet
- One Standby router per subnet
- Load balancing achieved between subnets
- Not within a single subnet

VRRP (Virtual Router Redundancy Protocol)

What VRRP?

VRRP is an open standard.

- It can run on devices from any vendor
- Cisco routers also support VRRP

In terms of functionality:

- VRRP is nearly identical to HSRP

VRRP Terminology:

VRRP uses different terms:

- Master → equivalent to HSRP Active
- Backup → equivalent to HSRP Standby

Only the terminology differs.

VRRP Multicast Address:

VRRP uses the IPv4 multicast address: 224.0.0.18

This address should be memorized for the exam.

VRRP Virtual MAC Address Format:

VRRP virtual MAC format:

0000.5e00.01XX

- XX = VRRP group number (hex)

VRRP MAC Example (Group 200):

- Decimal group number: 200
- Hexadecimal equivalent: c8

Virtual MAC:

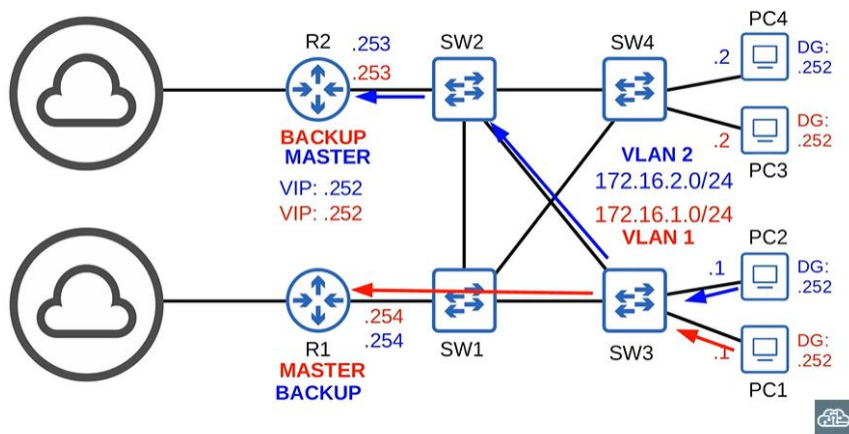
0000.5e00.01c8

VRRP Load Balancing:

Like HSRP:

- VRRP cannot load balance within one subnet
- VRRP can load balance between subnets

Different subnets can use different Master routers.



Relationship Between VLANs and Subnets:

- VLANs divide the network at Layer 2
- Subnets divide the network at Layer 3

They work together:

- One subnet per VLAN

GLBP (Gateway Load Balancing Protocol)

What GLBP?

GLBP is Cisco proprietary, like HSRP.

Key difference:

- GLBP load balances within a single subnet

GLBP Load Balancing Concept:

In GLBP:

- Hosts in the same subnet can use different routers as their default gateway

Example:

- PC1 uses R1
- PC2 uses R2
- Both are in the same subnet

GLBP Roles:

GLBP elects:

- AVG — Active Virtual Gateway
- AVF — Active Virtual Forwarders

Details:

- One AVG per subnet
- Up to four AVFs
- AVG can also act as an AVF

Each AVF:

- Acts as default gateway for a portion of hosts

GLBP Multicast and Virtual MAC

Multicast IP:

224.0.0.102

Virtual MAC format:

0007.b400.XXYY

- XX = GLBP group number
- YY = AVF number

Example:

- Group 1
- AVF 1

Virtual MAC:

0007.b400.0101

FHRP Comparison Chart

FHRP	Terminology	Multicast IP	Virtual MAC	Cisco proprietary?
HSRP	Active/Standby	v1: 224.0.0.2 v2: 224.0.0.102	v1: 0000.0c07.acXX v2: 0000.0c9f.fXXX	Yes
VRRP	Master/Backup	224.0.0.18	0000.5e00.01XX	No
GLBP	AVG / AVF	224.0.0.102	0007.b400.XXYY	Yes

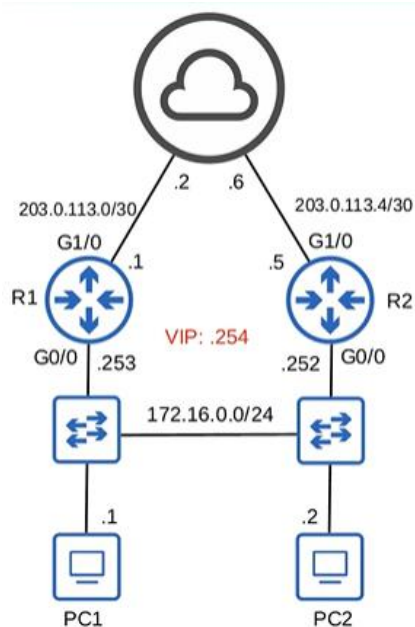
Basic HSRP Configuration

Lab Topology Overview

HSRP is configured on:

- R1 and R2
- Subnet: 172.16.0.0/24
- Virtual IP used: .254

All interface IPs are already configured.



HSRP Configuration on R1

HSRP is configured on the interface connected to the subnet.

Command base: standby

```
R1(config)#interface g0/0
R1(config-if)#standby ?
<0-255>      group number
authentication Authentication
bfd          Enable HSRP BFD
...          ...
```

Enabling HSRP Version 2:

```
R1(config-if)#standby version 2
R1(config-if)#standby ?
<0-4095>      group number
authentication Authentication
h5d          Enable HSRP BFD
```

After this: Group range becomes 0–4095

HSRP Group Selection:

- Group number chosen: **1**
- Matches VLAN number (not required, but preferred)
- Group number must match on both routers

```
R1(config-if)#standby 1 ?
authentication Authentication
follow      Name of HSRP group to follow
ip          Enable HSRP IPv4 and set the virtual IP address
ipv6        Enable HSRP IPv6
mac-address Virtual MAC address
name        Redundancy name string
preempt     Overthrow lower priority Active routers
priority    Priority level
timers      Hello and hold timers
track       Priority tracking
```

Configuring Virtual IP:

```
R1(config-if)#standby 1 ip 172.16.0.254
R1(config-if)#
```

Configuring Priority:

```
R1(config-if)#standby 1 priority ?
<0-255> Priority value

R1(config-if)#standby 1 priority 200
```

Priority determines Active router:

1. Highest priority (default 100)
2. Highest IP address (if priorities match)

Enabling Preemption:

```
R1(config-if)#standby 1 preempt
```

Preemption allows a router to:

- Take back the Active role after recovery
- Requires higher priority or IP address

HSRP Configuration on R2

- HSRP version 2 must match R1
- Same virtual IP is configured

```
R2(config-if)#standby version 2
R2(config-if)#
R2(config-if)#standby 1 ip 172.16.0.254
R2(config-if)#
R2(config-if)#standby 1 priority 50
R2(config-if)#
R2(config-if)#standby 1 preempt
```

Verifying HSRP with SHOW STANDBY

```
R1#show standby
GigabitEthernet0/0 - Group 1 (version 2)
  State is Active
    2 state changes, last state change 00:16:30
  Virtual IP address is 172.16.0.254
  Active virtual MAC address is 0000.0c9f.f001
    local virtual MAC address is 0000.0c9f.f001 (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.536 secs
  Preemption enabled
  Active router is local
  Standby router is 172.16.0.252, priority 50 (expires in 9.280 sec)
  Priority 200 (configured 200)
  Group name is "hsrp-Gi0/0-1" (default)
R1#
```

Final Review

Covered in this video:

- Purpose of FHRPs
- HSRP, VRRP, and GLBP overview
- Key multicast IPs and virtual MAC formats
- HSRP basic configuration commands

Commands reviewed:

- standby version 2
- standby ip
- standby priority
- standby preempt